

Lulu(Siwen) Shao

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RESEARCH INTEREST

Physically Grounded World Models, Spatial Intelligence, Cognitively Inspired Machine Learning Systems, AI for Scientific Simulation

EDUCATION

University of California, San Diego

B.S. Computer Science, Minor in Mathematics, GPA 3.9 Provost Honors

San Diego, CA

Sept 2023 - June 2027

PUBLICATIONS

(* indicates joint first authors)

- **[3DV'27 under review]** Haodong Li*, **Lulu Shao***, Haolin Lu*, Yu Fu, Yen-Ru Chen, Seemadhar Jain, Manmohan Chandraker. " *ϕ -Scene: Physically Grounded Image-to-3D Scene Reconstruction*" [[arXiv](#)] [[Page](#)]
- **[NeurIPS'26]** Hanan Gani, **Lulu Shao**, and Manmohan Chandraker. "*Mental Models for Multi-agent Agents*"
- **[NonAR-LM @ COLM'26]** Meliksah Yorulmazlar, Justice Sefas, Seunghyun Yoo, **Lulu Shao**. "*Feature Evolution in Diffusion Language Models*"

EXPERIENCE

UC San Diego - Computer Science and Engineering Departments

Undergraduate Researcher

Sept 2024 – Present

San Diego, CA

- Investigating how spatial intelligence models acquire affordance representations from physical interaction, with a focus on linking 3D scene structure, object relations, and action-relevant properties such as support, containment, and reachability.
- Contributed to ϕ -Scene, an image-to-3D reconstruction pipeline using SDF-based optimization and rigid-body simulation, by developing evaluation pipelines, running 3D scene benchmark comparisons, designing human/VLM-based perceptual evaluations, and writing main-paper and supplementary evaluation sections. (*under review*)
- Contributed to *Mental Models for Multi-Agent System*, a framework for improving open-source LLM/VLM agents on social interaction, negotiation, and Theory-of-Mind benchmarks using recursive ToM latents and belief-conditioned reward modeling. Analyzed learned mental-state latent representations through dialogue-level emotion clustering and documented ToM-related datasets and augmentations for mental-state supervision. (*NeurIPS'26*)
- Developed a benchmark dataset for novel view synthesis under diverse lighting, material, and trajectory conditions, enabling standardized evaluation across inverse rendering models, and automated evaluation pipelines supporting robust comparisons across NeRF, 3DGS, InstantNGP, and 10+ related inverse rendering methods. Mentored by professors Tzu-Mao Li and Ravi Ramamoorthi through the Early Research Scholars Program

Google

Software Engineer Intern - Google TV LauncherX Team

Jun 2026 – Sept 2026

San Jose, CA

- Developed full-stack "From Your Services" topic page using Kotlin client stream-stubs, Java backend preprocessors, and Protobufs, enabling targeted content discovery for millions of users with zero reload UI updates.
- Authored technical design documents and steered architecture decisions across global teams, successfully averting high risk system migration and accelerating project delivery.

Amazon Web Services

Software Development Engineer Intern - Database Migration Service - Platform team

Jun 2025 – Sept 2025

Seattle, WA

- Developed an AWS Level 3 CDK construct to monitor cloud resource utilization with automated CloudWatch service quota alarms and quota increase requests, released to production to support Sev2-Sev5 event monitoring.
- Built Lambda and EventBridge workflows deployed through CloudFormation to track resource utilization metrics.

Verizon

Machine Learning Engineer Intern (AI Studio)

Aug 2024 – Dec 2024

Hybrid, UCLA, CA

- Designed and created a dual-model computer vision/segmentation pipeline using YOLOv8 and OpenCV masking to match phone cases to devices in real time, achieving 97%+ accuracy generalizing across 600+ diverse retail image datasets and enhancing retail operation efficiency.

Engineers For Exploration

Oct 2023 – Dec 2024

Researcher (Collaborated with San Diego Zoo Institute for Conservation Research)

UCSD, San Diego, CA

- Deployed machine learning pipelines that pull from unsupervised and supervised methods in conjunction with digital signal processing techniques to extrapolate meaningful data from Peruvian Amazon audio.
- Improved species monitoring by developing a pipeline to accurately identify various species vocalizations.

PROJECTS

Deep Learning for Climate Emulation

March 2025 – June 2025

- Developed and fine-tuned a Vision Transformer (ViT) model to forecast global surface temperature and precipitation from emissions and solar radiation data, reducing loss score from 3.928 to 0.8297 by capturing long-range spatial dependencies and optimizing hyperparameters via Optuna.
- Experimented with and benchmarked multiple deep learning architectures—including CNNs, U-Nets, Spherical Fourier Neural Operators (SFNO), and spatiotemporal Transformers—to evaluate trade-offs in spatial vs. temporal modeling and guide architecture selection.

TEACHING

UC San Diego - Computer Science and Engineering Departments

Sept 2024 - Present

Undergraduate Teaching Assistant

San Diego, CA

Supported and mentored students across introductory programming, systems, AI, deep learning, and large-codebase software engineering courses by leading office hours, discussion/study sessions, rubric/assignment design, and grading.

- Fall 2026 - CSE 190: Working with Large Codebases: Traditional and Agentic Software Engineering under Professor Gerald Soosairaj
- Spring 2026 - CSE 151B: Deep Learning under Professor Rose Yu
- Winter 2026 - CSE 25: Introduction to Artificial Intelligence under Professor Trevor Bonjour
- Fall 2025 - CSE 8A: Introduction to Programming and Computational Problem-Solving under Professor Gerald Soosairaj
- Spring 2025 - CSE 29: Systems Programming and Software Tools under Professor Gerald Soosairaj
- Fall 2024 - CSE 8A: Introduction to Programming and Computational Problem-Solving under Professor Gerald Soosairaj

TECHNICAL SKILLS

Languages: Java, Python, C, C++, JavaScript, HTML, CSS, R, Shell Script, MATLAB

Libraries: PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, NumPy, OpenCV, OpenCL, OpenGL, React

Tools: MySQL, ROS, Docker, Figma, JUnit, Git, Vim, Roboflow, Research design, Project management

Research: Literature review, experimental design, benchmark construction, ablation studies, model evaluation, human evaluation, VLM-based evaluation, representation analysis, paper writing, LaTeX/Overleaf.